LOGISTICS OPERATIONS & PERFORMANCE ANALYSIS

PROJECT REPORT

1. PROJECT OVERVIEW

This project analyses logistics operations data to evaluate operational performance, fleet utilisation, driver performance, fuel and maintenance costs, safety performance, customer revenue, and route profitability using Power BI.

2. DATA PREPARATION & ANALYSIS

The datasets were cleaned and transformed using Power Query before analysis. A relational data model was developed in Power BI, and DAX measures were created to calculate key performance indicators. The analysis was presented through interactive dashboards covering logistics operations, fleet performance, driver and safety performance, and customer and route profitability.

3. KEY FINDINGS

3.1 Logistics Operations Overview

1. Overall logistics performance:
   The operation generated 262.53M in total revenue from approximately 85K loads, with total fuel costs of 95.59M.

2. On-time delivery performance:
   The overall on-time delivery rate was 55.67%, indicating that a significant proportion of deliveries were not completed on time.

3. Monthly revenue and load volume:

   Monthly revenue and load volume fluctuated throughout the period from 2022 to 2024, with noticeable variations across different months rather than a consistent upward or downward trend.

4. Monthly on-time delivery performance:

   On-time delivery rates varied throughout the year, reaching their lowest point around March and their highest level around October, showing fluctuations in delivery performance across months.

5. Load type distribution:

   Refrigerated loads accounted for a slightly larger volume than Dry Van loads, making them the more prevalent load type in the dataset.

6. Monthly fuel costs:

   Fuel costs fluctuated considerably across the months, with February showing one of the lowest levels, while higher fuel-cost levels were observed around March, July/August, and October.

3.2 Fleet, Fuel & Maintenance

1. Fleet utilisation:

   The fleet recorded an average utilisation rate of 83.0%, with monthly utilisation fluctuating between approximately 80% and 87% during the period.

2. Fleet size and mileage:

   The analysis covered 120 trucks, with an average of 1.31M total miles per truck as displayed on the dashboard.

3. Fuel efficiency:

   The overall average fuel efficiency was 6.50 MPG, with the trucks shown on the dashboard recording relatively similar MPG levels.

4. Maintenance costs:

   Total maintenance cost was 5.73M. Maintenance cost per mile varied across the trucks shown, with the highest values around 0.058 per mile and the lowest around 0.050 per mile.

5. Monthly fuel costs:

   Monthly fuel costs fluctuated throughout the year, with lower costs around February and September, while higher costs occurred around July/August and October.

3.3 Drivers & Safety Performance

1. On-time delivery performance:

   The average on-time delivery rate across drivers was 44.60%, indicating room for improvement in delivery reliability.

2. Driver population:

   The analysis covered 124 drivers, providing a broad view of individual driver performance.

3. Safety incidents:

   A total of 170 safety incidents were recorded, of which 64 were classified as preventable incidents.

4. Driver revenue performance:

   Average revenue per mile was 2.15, providing an indication of the revenue generated relative to distance traveled.

5. Driver performance variation:

   The average on-time delivery rate varied across drivers, with the highest-performing drivers reaching approximately 50%, while the lowest shown were around 47%.

6. Fuel efficiency among top drivers:

   The dashboard identifies the top 10 drivers by average MPG, with the displayed drivers recording relatively similar fuel-efficiency levels.

7. Safety performance varied by driver:

   Safety incidents and preventable incidents were not evenly distributed across drivers, with some drivers recording noticeably higher incident counts than others.

3.4 Customer & Route Profitability

1. Customer and route coverage:

   The analysis covered 200 customers across 58 routes, with an average revenue per load of 3.07K and an average base rate per mile of 2.20.

2. Customer revenue concentration:

   Revenue was concentrated among a group of top customers, with First Group recording the highest revenue among the top 10 customers displayed.

3. Revenue by booking type:

   Dedicated bookings generated the highest share of revenue at 130.24M (49.61%), followed by Contract bookings at 66.32M (25.26%) and Spot bookings at 65.97M (25.13%).

4. Loads by customer type:

   Dedicated loads represented the largest share of total loads at approximately 32K (37.71%), followed by Spot loads at 27K (31.44%) and Contract loads at 26K (30.85%).

5. Route profitability:

   Profit after fuel cost varied across the top 10 routes, with the leading routes generating approximately 8–9M in profit after fuel costs, while the lower-performing routes generated approximately 7–8M.

4. BUSINESS INSIGHTS & RECOMMENDATIONS

4.1 Revenue is Relatively Fuel-Intensive

Fuel costs account for a substantial portion of the revenue generated by the operation. This makes fuel efficiency an important driver of overall profitability, particularly because fuel costs fluctuate considerably across months.

Recommendation: Rather than looking at fuel cost in isolation, management should monitor fuel cost relative to revenue, mileage, and truck performance to identify periods and vehicles where fuel consumption is disproportionately high.

4.2 Dedicated Bookings Appear to Generate Stronger Revenue Relative to Load Volume

Dedicated bookings generated 49.61% of revenue while accounting for 37.71% of loads. This means their contribution to revenue is greater than their share of total load volume.

Recommendation: The business should investigate what makes dedicated bookings more valuable, such as pricing, mileage, customer characteristics, or contract terms and determine whether similar characteristics can be used to improve the economics of other booking types.

4.3 Contract and Spot Loads Show a Different Revenue Pattern

Contract and spot bookings each accounted for roughly 31% of load volume, but each contributed only about 25% of revenue. Their revenue contribution therefore does not match their share of operational volume.

Recommendation: Examine the pricing and cost structure of these bookings to determine whether certain contracts, routes, or customers are generating relatively low returns despite requiring comparable operational capacity.

4.4 Delivery Performance Varies Meaningfully Across Time

The monthly on-time delivery trend does not remain consistent throughout the year. There are noticeable periods of stronger and weaker performance, suggesting that delivery reliability may be influenced by operational conditions that change over time.

Recommendation: Compare the characteristics of stronger-performing periods with weaker ones such as routes, drivers, load types, and fleet availability to identify practices that could be replicated during weaker periods.

4.5 Preventable Incidents Represent a Significant Portion of Safety Incidents

Of the 170 safety incidents recorded, 64 were preventable, meaning approximately 38% could potentially have been avoided.

Recommendation: Instead of treating all safety incidents equally, management could focus first on the drivers, routes, or operating conditions associated with the preventable incidents. This would allow safety interventions to be more targeted.

4.6 Fleet Utilisation is Reasonably Strong, but Consistency Matters

The fleet recorded an average utilisation rate of 83%, but the monthly trend shows noticeable fluctuations. The issue is therefore not simply whether the fleet is being utilised, but when and why utilisation falls.

Recommendation: Investigate the periods of lower utilisation alongside load volume and fleet availability. This could reveal opportunities to improve vehicle scheduling without necessarily increasing fleet size.

4.7 Route Profitability should be Considered after Fuel Costs

The route analysis shows that the most profitable routes are not necessarily being evaluated on revenue alone; profitability after fuel cost provides a more useful view of which routes actually contribute value.

Recommendation: Use profit after fuel cost per route as a recurring performance metric and investigate routes that consistently underperform despite carrying significant load volumes.

4.8 Customer Concentration Creates an Opportunity for Deeper Analysis

Revenue is not distributed evenly across customers, with some customers contributing considerably more revenue than others.

Recommendation: The business should examine its highest-value customers alongside their booking types, routes, and profitability not just revenue to determine which customer relationships are genuinely the most valuable.

5. PROJECT OUTCOME

The project resulted in an interactive Power BI dashboard that brings together logistics, fleet, fuel, maintenance, driver, safety, customer, and route performance in a single reporting solution. The analysis provides a structured view of operational performance and highlights areas where further investigation can support efficiency, profitability, and service improvement.